

What is a React Component?

A **React Component** is an independent, reusable piece of UI.

Components divide a webpage into smaller reusable parts.

Example:

- Header
- Footer
- Navigation Bar
- Home Page
- Contact Page

Components vs JavaScript Functions

React Component	JavaScript Function
Returns JSX	Returns any JavaScript value
Used for UI	Used for logic
First letter should be Capital	Any valid name
Can use React features	Cannot use React features unless used as component

Types of Components

1. Functional Component

Uses JavaScript functions.

Example:

```
function Home() {  
  return <h1>Home</h1>;  
}
```

2. Class Component

Uses ES6 classes.

Example:

```
class Home extends React.Component {  
  render() {  
    return <h1>Home</h1>;  
  }  
}
```

Class Component

A class component

- extends React.Component
- contains a render() method
- returns JSX

Constructor

The constructor initializes state and binds methods.

Example

```
constructor(props) {  
  super(props);  
}
```

render()

render() returns the JSX that should appear on the screen.

```
render() {  
  return(  
    <h1>Hello</h1>  
  );  
}
```

Practical:

Home.js:

```
import React, { Component } from "react";
```

```
class Home extends Component {
```

```
  render() {
```

```
    return (
```

```
      <div>
```

```
        <h2>Welcome to the Home page of Student Management Portal</h2>
```

```
      </div>
```

```
    );
```

```
  }
```

```
}
```

```
export default Home;
```

About.js:

```
import React, { Component } from "react";
```

```
class About extends Component {
```

```
  render() {
```

```
    return (
```

```
      <div>
```

```
        <h2>Welcome to the About page of the Student Management Portal</h2>
```

```
      </div>
```

```
    );
```

```
  }
```

```
}
```

```
export default About;
```

Contact.js:

```
import React, { Component } from "react";
```

```
class Contact extends Component {
```

```
  render() {
```

```
    return (
```

```
      <div>
```

```
        <h2>Welcome to the Contact page of the Student Management Portal</h2>
```

```
      </div>
```

```
    );
```

```
  }
```

```
}
```

```
export default Contact;
```

App.js:

```
import React from "react";
```

```
import Home from "../Components/Home";
```

```
import About from "../Components/About";
```

```
import Contact from "../Components/Contact";
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
    <Home />

    <About />

    <Contact />

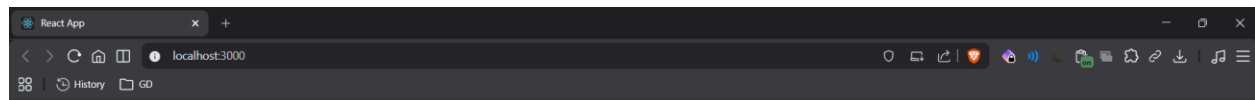
  </div>

);

}
```

export default App;

Output:



Welcome to the Home page of Student Management Portal

Welcome to the About page of the Student Management Portal

Welcome to the Contact page of the Student Management Portal